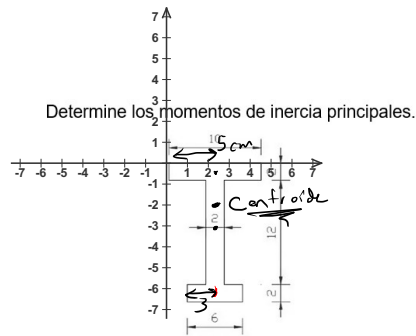


Centroid de y

$$\frac{(5)(2) + (5)(20) + (5)(24)}{56}$$

= 4.85 cm ←

$$I_x = I_{x_0} + A y^2$$

$$I_{x_1} = \frac{1}{12} (6)(2)^3 + (2)(15)^2 = 2704$$

$$I_{x_2} = \frac{1}{12} (10)(2)^3 + (2)(1)^2 = 26.66$$

$$I_{x_3} = \frac{1}{12} (12)(2)^3 + (2)(8)^2 = \frac{1824}{4554.66}$$

$$I_y = I_{y_0} + A x^2$$

$$I_{y_1} = \frac{1}{12} (6)(2)^3 + (2)(5)^2 = 336$$

$$I_{y_2} = \frac{1}{12} (10)(2)^3 + (2)(5)^2 = 666.6$$

$$I_{y_3} = \frac{1}{12} (12)(2)^3 + (2)(5)^2 = \frac{608}{1610.6} \checkmark$$

$$I_{x_y} = 0 + (5)(15)(12) = 900$$

$$I_{x_y_2} = 0 + (5)(1)(20) = 100$$

$$I_{x_y_3} = 0 + (5)(8)(24) = \frac{960}{1960}$$

$$A_1 = 6 \times 2 = 6 \times h = 12$$

$$A_2 = 10 \times 2 = 20$$

$$A_3 = 12 \times 2 = 24$$

Centroide en x

$$\frac{(5)(12) + (5)(20) + (5)(24)}{56}$$

$$\rightarrow = 5 \text{ cm}$$

$$4554.66 - (56)(4.85)^2$$

$$= 3237.4$$

$$1610.6 - (56)(5)^2$$

$$= 210.6$$

$$1960 - (56)(4.85)(5) = 602 \checkmark$$

$$C = \frac{3237.4 + 210.6}{2} = 1724 \checkmark$$

$$R = \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + (I_{x_y})^2} = \sqrt{\left(\frac{3237.4 - 210.6}{2}\right)^2 + (602)^2} = 1628.73 \checkmark$$

$$I_{\max} = 1724 + 1628.73 = 3352.73 \checkmark$$

$$I_{\min} = 1724 - 1628.73 = 95.27 \checkmark$$

$$\tan 2\theta_p = \frac{2I_{x_y}}{I_y - I_x} = \frac{2(602)}{210.6 - 3237.4} = -0.397$$

$$2\theta_p = -21.65$$

$$\theta_p = -10.82$$

